Programming Techniques – Using C Language

Pointers

Objective:
The purpose of this lab is to get an idea about what are pointers and how they work.

1. Type and execute the following program to get an idea about the memory addresses of the variables.

```
#include <stdio.h>
main(){
        int j,k;
        printf("\n");
        printf("j is stored at %p\n" ,&j);
        printf("K is stored at %p\n", &k);
        return 0;
}
```

2. Sizes of the data types in bytes.
   Execute the given program which gives the size of the data type (short) in bytes.

```
#include <stdio.h>
int main(){
        printf("size of a short is %d\n", sizeof(short));
        return 0;
}
```

- Modify this program to output the sizes of the integer, long, character and float data types.

3. Execute the following program which is a simple example to understand dereferencing in C pointers.

```
#include <stdio.h>
int main() {
        int i;
        int *p;  /* a pointer to an integer */
        p = &i;
        *p=5;
        printf("%d %d\n", i, *p);
        return 0;
}
```

4. Type and execute the given program and get an idea about the pointer basics.

```
#include <stdio.h>
int main(){
        int j,k;
        int *ptr;
        j=1;
        k=2;
        ptr=&k;
        printf("\n");
        printf("j has the value %d and is stored at %p\n" ,j, &j);
        printf("K has the value %d and is stored at %p\n" ,k, &k);
        printf("ptr has the value %p and is stored at %p\n" ,ptr, &ptr);
        printf("the value of the integer pointed to by ptr is %d \n" ,*ptr);
        return 0;
}
```

- Modify this program by adding the following statement before the first printf statement and analyze the output.

***ptr=7;**

5. Following example is to understand how to change the value of a variable using pointer.

```
#include <stdio.h>
int main(){
        int x, *p;
        p = &x;         /* initialise pointer */
        *p = 0;         /* set x to zero */
        printf("x is %d\n", x);
        printf("*p is %d\n", *p);
        *p += 1;        /* increment what p points to */
        printf("x is %d\n", x);
        (*p)++;         /* increment what p points to */
        printf("x is %d\n", x);
        return 0;
}
```

6. Write a C program to do the followings. Introduce integer variables **a, b** and integer pointer variables **p, q.** Assign **a** to **2, b** to **8**, **p** to the address of **a**, and **q** to the address of **b**. Then print the following information:

    - The address of a and the value of a.
    - The value of p and the value of *p.
    - The address of b and the value of b.
    - The value of q and the value of *q.
    - The address of p.
    - The address of q.

7. Build a c program according to the following guidelines.

    - Declare integer, floating point and character variables called num, fl and ch respectively.
    - Declare integer, floating point and character pointer variables called numptr, flptr and chptr respectively.
    - Assign values 234, 10.345, ‘a’ to variables num, fl and ch respectively. Assign address of num to numptr.
    - Assign address of fl to flptr.
    - Assign address of ch to chptr.
    - Print the values of *numptr, numptr and &numptr.
    - Print the values of *flptr, flptr and &flptr.
    - Print the values of *chptr, chptr and &chptr.
    - Now add (*chptr)++; statement to your program and display the value of your variable ch.
    - Do this to other two pointer variables and display the values of num and fl.